

Inside SeNMo

Multi-omics survival modeling on the TCGA pan-cancer atlas — from first-principles data exploration to a reproduced, HPC-scale preprocessing pipeline.

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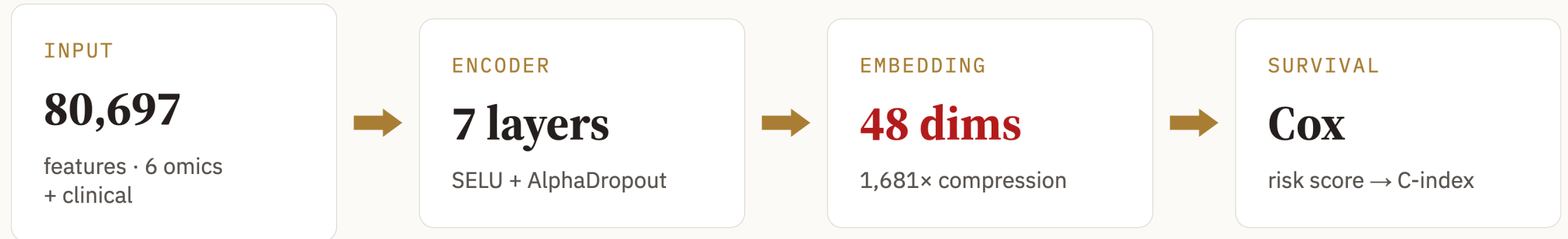
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THE MODEL

One model, one bottleneck

SeNMo is a self-normalizing multi-omics network: it compresses every patient's molecular profile through a narrow encoder and predicts overall survival with a Cox objective.



Self-normalizing activations (SELU + Alpha-Dropout) hold the signal stable through depth without batch-norm — which matters because the six omics live on wildly different numeric scales.

THE THESIS UNDER TEST

Age and sex should be *first-class variables* in a survival model — not one diluted column among 80,697 features.

Ovarian cancer (anatomically all-female) is the entry point; the broader claim is that demographic biology belongs in the representation itself, not averaged away by a sex-pooled cohort.

THE ARC

Seven weeks — and then the cluster

WEEK 1

Foundations

Architecture & the hypothesis

WEEK 2

TCGA-OV

Cohort exploration & survival stats

WEEK 3

ML bridge

Every concept mapped to code

WEEK 4

Replication

Reproduce Fig 7 + probe the embedding

WEEK 5

Cross-cancer

BRCA, LUAD & the female pipeline

WEEK 6

The pivot

Demographics as first-class

WEEK 7

To the server

HPC access on Cayuga

WEEK 8 +

On the cluster

Reproducing preprocessing at scale

Reading the data before the model

The first weeks were an apprenticeship in survival analysis from first principles — Kaplan–Meier curves, the Cox proportional-hazards model, and the concordance index, built up rather than called blindly.

Censoring is the crux: a patient alive at last contact is kept as partial information, not discarded. The **C-index** is survival's analogue of AUC — 0.5 is a coin flip, 1.0 is perfect.

428 women

the TCGA-OV cohort — high-grade serous ovarian carcinoma

C-index 0.626

Cox on 3 clinical variables — the performance floor the embedding must beat

r 0.75 vs 0.10

PTEN vs BRCA2 mRNA↔protein — why one omic cannot substitute for another

Reproducing the paper's Figure 7

A fresh 5-fold cross-validated Cox model on the frozen embeddings, per cancer. Reproducing the ranking and the point estimates is what earns the right to retrain.



Bars scaled from 0.50 (chance) to 0.95. Point estimates matched the published figure closely — clearing the model for everything that followed.

What does the 48-dim embedding encode?

ABLATION

Redundant across cancers, not within

Pan-cancer C-index holds at **0.79 even after zeroing 90%** of the dimensions — the easy "which cancer" signal is stored many times over. Ranking survival *inside* one cancer is far less redundant and decays toward chance.

AGE & SEX PROBE

A faint, non-linear residue

Demographics are recoverable, but only non-linearly: **sex AUC 0.737**, age MAE 9.6 yr (r 0.51) with gradient boosting — while linear probes plateau near 0.63. You only learn the signal is non-linear by putting both side by side.

SeNMo discards age

0.605

age alone (Cox C-index, OV)

0.575

the full 48-dim embedding

+0.045

gain from adding age back (CI
excludes 0)

Age is nearly **unrecoverable from the embedding ($r \approx -0.07$)** — even though age was in SeNMo's clinical input. The 80k→48 bottleneck squeezes it out. Making demographics first-class means recovering prognostic signal the model actively drops.

Glioblastoma, the showpiece: the embedding predicts survival at chance (0.49) while a single age variable reaches 0.65.

THE CONVERGED NARRATIVE

Three lines of evidence, one conclusion

L1

Age is dropped

The embedding loses age across nearly every cancer; recovering it adds +0.045.

L2

Recovery is additive

Conditioning (FiLM) doesn't beat concatenation — a head-level patch is trivial. The real fix is upstream.

LUAD

Filtering buys nothing

Female-only (0.675) $\approx$ mixed (0.679) on the same held-out women. Male data is neither noise nor much signal.

THE CONCLUSION

The lever is *demographic recovery* — retrain the encoder with age & sex first-class, not sex-filtering.

WEEK 7

To the server

Every result so far was a frozen-embedding probe — possible on a laptop precisely because the encoder itself cannot be trained there. That needs a GPU cluster. Week 7 is the transition from analysis to infrastructure.



Cornell CAC "Cayuga" cluster · Slurm scheduler · Xena ETL for five omics + phenotype + survival · GDC MAFs for mutations.

CHAPTER 8 · ON THE CLUSTER

Reproducing the preprocessing at pan-cancer scale

The work after the written portfolio: rebuilding SeNMo's six-omic pipeline across all 33 TCGA cohorts on Cayuga — faithfully, and fast enough to actually run.

RESULTS

81,240 features — within 0.7% of the authors

The assembled matrix is **11,127 patients × 81,240 features**. Two omics match to the exact feature; every omic is within 2.4%.

Omic	Authors	Ours	Δ%
Clinical	4	4	0.0
Protein (RPPA)	472	472	0.0
miRNA	1,730	1,746	+0.9
Gene expression	8,794	8,582	−2.4
DNA methylation	52,396	53,387	+1.9
DNA mutation	17,301	17,078	−1.3
6-MODAL TOTAL	80,697	81,240	+0.7

Residual gaps live only in the **tunable, data-dependent filters** — clinical and protein, which have no such filter, match exactly.

ENGINEERING

Making it run — and making it correct

PARALLELISM

Slurm job arrays

All 33 cohorts processed in parallel, each picking its cohort from the array index — a multi-day serial job down to hours.

MEMORY

Parquet checkpoints

Wide CSV writes on ~130,000-column matrices ran out of memory and died. Parquet writes the same data in seconds.

CORRECTNESS

A 6,197-sample bug

A survival-label join silently duplicated 6,197 samples. Diagnosed, fixed by holding labels raw — duplicates → 0.

The statistical operations are the authors' code verbatim; only the execution harness — array indexing, Parquet, absolute paths — is adapted to run headless at scale.

VALIDATION

Checking against the authors' patients

The authors released only their **combined** embedding set, never their train/test split — so a patient-by-patient train match is impossible. The honest check is **containment**: is every patient I kept one the authors also had?

It holds. The patients I'm missing are precisely their held-out **test split** — expected, since I built only the training side.

Authors (train + test)

10,938

Mine (training matrix)

9,248

Shared patients

9,151

Authors-only 1,787 (their test split) · mine-only 97 (~1%, under review) · Jaccard 0.83.

OUTCOMES

What the work built

Skills

- Survival analysis — Cox PH, C-index, Kaplan–Meier
- Multi-omics & TCGA / UCSC Xena data
- HPC — Slurm job arrays, batch workflows
- Reproducible, auditable pipelines
- Python · pandas · PyTorch · scikit-survival

Delivered

- Reproduced the paper's Figure 7
- Reproduced full preprocessing within 0.7%
- Found SeNMo systematically drops age
- Caught & fixed a 6,197-sample bug
- Sharpened the thesis with evidence

SEVEN WEEKS — AND THE CLUSTER

From a Kaplan–Meier curve to a pan-cancer pipeline.

An apprenticeship, a reproduction, a genuine finding, and the infrastructure to build on — a defensible, evidence-driven redirection of a foundation-model project.

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